

DIgSILENT *PowerFactory*^{*}

1 General Description

The Nine-bus System was introduced in the book *Power System Control and Stability* by P. M. Anderson and A. A. Fouad [1]. It represents a small transmission system which consists of 9 buses (nodes), 3 generators, 3 loads, 6 lines and 3 transformers, the single line diagram is shown in Figure 1.



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2 Model Parameters

The nominal voltage of the transmission system is 230 kV, the nominal frequency is 60 Hz. The following subsections describe the parameters of the elements as used for balanced load flow calculation and RMS simulation. Data have been taken from [1].

2.1 Loads

During load flow calculation, the loads of the Nine-bus System have constant active and reactive power demand [1], they are not voltage-dependent. This is achieved by *disabling* the load option “Consider Voltage Dependency of Loads” in the *PowerFactory* load flow calculation command.

Load data (active power P and reactive power Q) are listed in Table 1.

The steady-state load flow determines the initial values for the stability simulation (dynamic RMS phasor simulation). During RMS simulation the loads are considered as equivalent impedances.

2.2 Generators

Generator “G1” is the slack machine, voltage 1.04 p.u. and 0 degrees. The other generators are configured to control the active power injection and voltage magnitudes at the connected buses, therefore the active power dispatch and controlled voltage magnitudes at their terminals are given.

The data have been taken from [1] and are listed in Table 2 and 4.

The reactances x of the generators have been adapted to the generator rated power $S_{r,gen}$ using Equation 1. The inertia time constant H based on the rated active power $P_{r,gen}$ has been calculated from the stored energy E at nominal speed with Equation 2. The results are presented in Table 3.

$$\begin{aligned} x[\text{p.u. generator base}] & \\ = x[\text{p.u. system base}] \cdot \frac{S_{r,gen}[\text{MVA}]}{100 \text{ MVA}} \end{aligned} \quad (1)$$

$$H = \frac{E}{P_{r,gen}} \quad (2)$$

For RMS simulation, four dynamic models are available for synchronous generators in *PowerFactory 2016*: a standard model, a classical model, a 3.3 model and a model for asynchronous starting of a synchronous machine.

- The *standard model* represents a field winding in the d-axis, and a damper winding in the d- and q-axis [2].
- The *classical model* is a simplified model consisting of a voltage source behind an impedance [2].
- The *3.3 model* contains a field winding in the d-axis, two damper windings in the d-axis and three damper windings in the q-axis [2].
- The *model for asynchronous starting* contains additional impedance branches in the internal equivalent circuit which are relevant during asynchronous operation of a synchronous machine [2].

To reproduce the examples described in [1], the standard model and the classical model are used in the Nine-bus System in accordance to [1].

2.3 Transmission Lines

Line data are given in per unit (p.u.) on a $S_b = 100 \text{ MVA}$ system base as represented in Table 6 [1]. As there is no line length given in [1], the length of each line in the *PowerFactory* model has been set to 1 km.

For the *PowerFactory* model input data are required in Ω/km and $\mu\text{F}/\text{km}$ respectively. Line data have been recalculated for the network model with the nominal voltage $U_n = 230 \text{ kV}$ using Equations (3) – (5).

$$R [\Omega] = r [\text{p.u.}] \cdot \frac{U_n^2 [\text{kV}^2]}{S_b [\text{MVA}]} \quad (3)$$

$$X [\Omega] = x [\text{p.u.}] \cdot \frac{U_n^2 [\text{kV}^2]}{S_b [\text{MVA}]} \quad (4)$$

$$B [\mu\text{S}] = b [\text{p.u.}] \cdot \frac{S_b [\text{MVA}]}{U_n^2 [\text{kV}^2]} \cdot 10^6 \quad (5)$$

Lines are assumed to be overhead lines and since the rated current of each line is not known, it is assumed to be 1 kA.

2.4 Transformers

Transformer data are given in per unit (p.u.) on a 100 MVA system base as represented in Table 7 [1]. In the *PowerFactory* model, the rated power of the transformers has been chosen according to the size of the connected generators. The reactances x of the transformers have been adapted to the transformer rated power $S_{r,trf}$ using Equation 6. Transformer parameters of the *PowerFactory* model are given in Table 8.

$$x[\text{p.u. transformer base}] = x[\text{p.u. system base}] \cdot \frac{S_{r,trf}[\text{MVA}]}{100 \text{ MVA}} \quad (6)$$

The vector group of all transformers has been assumed to be YNd5. This leads to an additional phase shift of 150 degrees for the voltage angles at the 230 kV level in the *PowerFactory* results compared to the results obtained in [1].

3 Load Flow Results

The steady-state load flow is examined by executing the load flow calculation (↻). The results of the *PowerFactory* load flow calculation are depicted in Figure 2 and additionally provided in Appendix B.

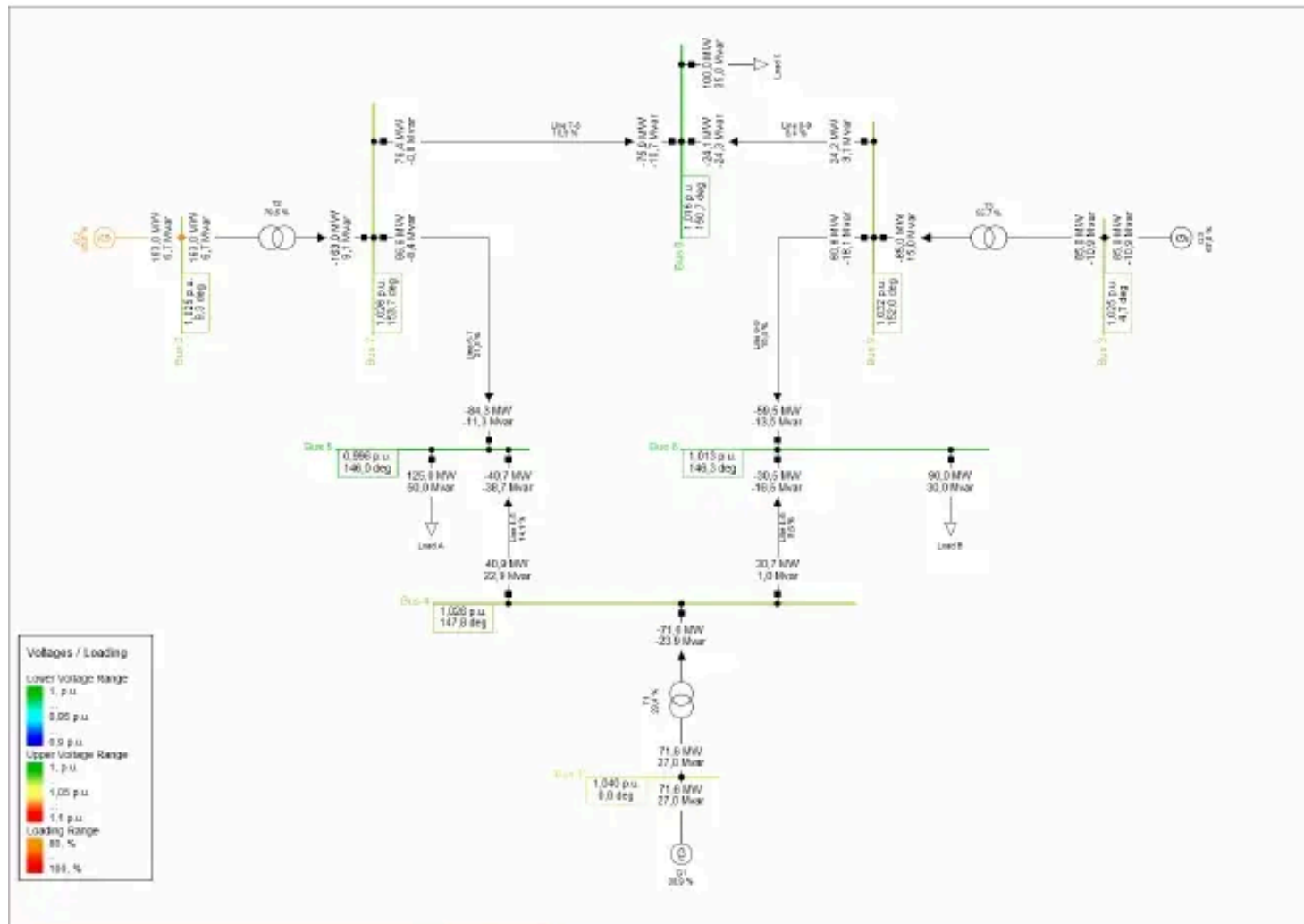


Figure 2: Results of the load flow calculation

4 RMS Simulations

A number of different RMS simulations is performed to analyse the transient stability of the Nine-bus System and the effect of different excitations systems and of a power system stabiliser.

4.1 Five Cycles Fault

In this study case the Example 2.7 of the book *Power System Control and Stability* [1] is reproduced. In this example the classical synchronous generator model is used as described in [1]. In order to reproduce the classical model described in the book (section 2.5.1) the transient reactance x'_d is used as stator reactance (x_{str}) in the *PowerFactory* model.

A three-phase short-circuit event is simulated at an end of the Line 5-7. The fault is cleared in five cycles (83.3 ms) by tripping the faulted line.

The resulting curves for the rotor angles of generators “G2” and “G3” with reference to “G1” are shown in Figure 3. The rotor angles of both generators reach a maximum value and then decrease. Transient stability of the system is given in this scenario. The results correspond with [1].

4.2 Impact of the Excitation System

This simulation described in Section 4.1 was carried out without any controller taken into account. However, in a real system the controls have a big impact on the stability of the system. In the following study cases with different types of excitation systems are presented. In these cases the standard model of the synchronous machine is used, which allows to connect an excitation system (automatic voltage regulator, AVR) to the generator.

The following types of the excitation system are modelled at the generator G2, in order to reproduce results provided in [1]:

- Standard model of the synchronous machine available in *PowerFactory* (no additional control)
- Standard model of the synchronous machine

with an AVR model IEEE type 1 Amplidyne

- Standard model of the synchronous machine with an AVR model IEEE type 1 Mag-A-Stat
- Standard model of the synchronous machine with an AVR model IEEE type 3 SCPT

The excitation models are taken from the global library available in *PowerFactory* and the parameter values are modified according to the data provided in [1]. A three-phase fault with a duration of three cycles (50 ms) is simulated at the end on the Line 5-7, the fault is cleared by tripping the faulted line. The results are depicted in Figure 4.

The results show good consistence with [1]. The maximum rotor angle differs depending on the AVR type used. This demonstrates the effect which the excitation system has on the transient stability of the generator. The smaller the maximum rotor angle is, the larger is the margin to the stability limit.

4.3 Impact of an Power System Stabiliser

In order to simulate the impact of the Power System Stabiliser (PSS), a PSS model as specified in [1] is added to the control of the machine. The PSS model is taken from the global library available in *PowerFactory* and the parameter values are chosen according to the data provided in [1]. The case with the AVR model IEEE type 1 Mag-A-Stat was used for this study. Results are shown in Figure 5 and Figure 6. The PSS damps the oscillation of the generator by influencing the excitation voltage.

References

- [1] P. Anderson and A. Fouad, *Power System Control and Stability*, 1st ed. Iowa State University Press, Ames, Iowa, U.S.A., 1977.
- [2] *DIgSILENT PowerFactory 2016: Technical Reference Documentation Synchronous Machine*, Version 2016, 1st ed., DIgSILENT GmbH, Heinrich-Hertz-Str. 9, 72810 Gomaringen, Germany, 2016.

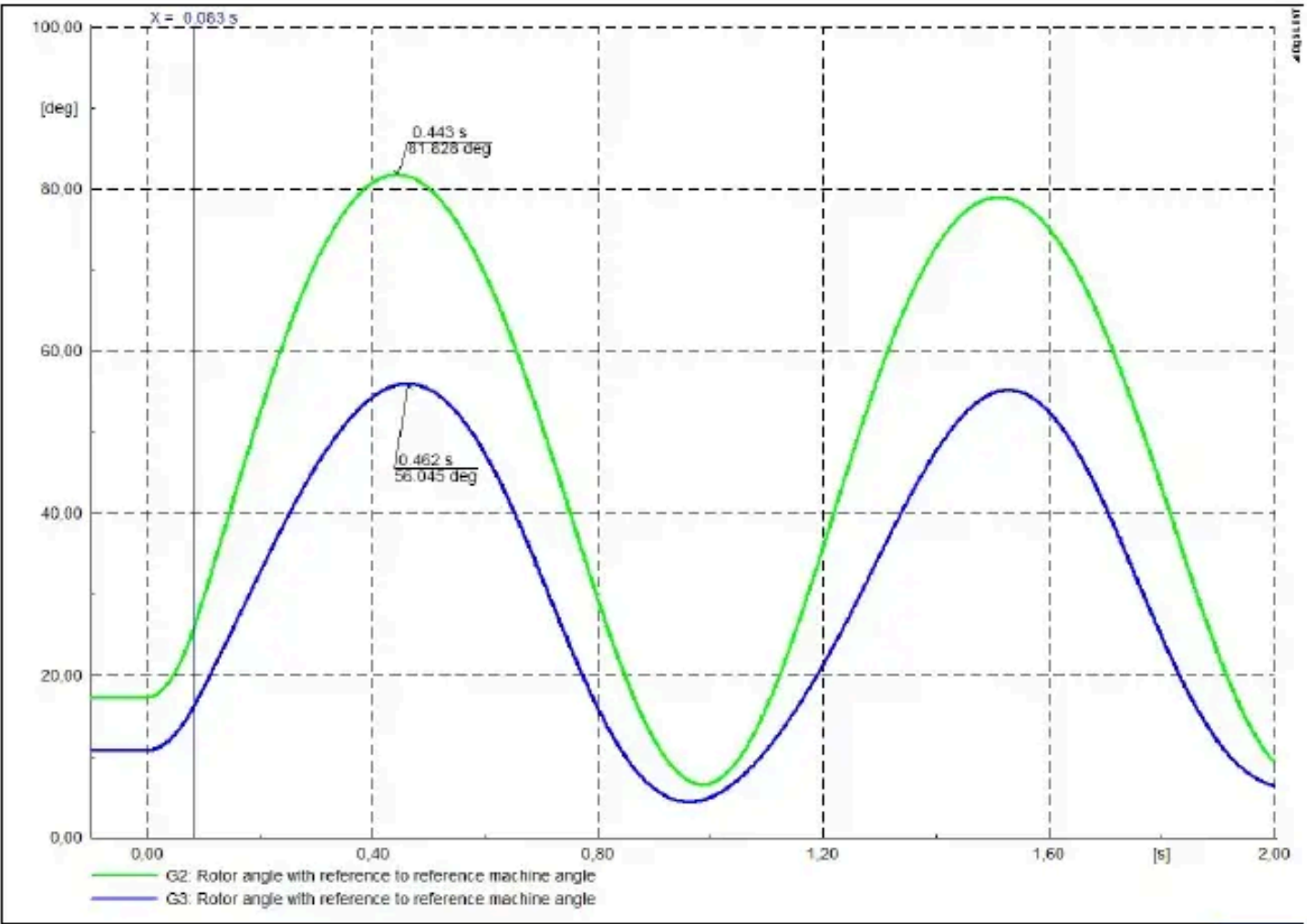


Figure 3: Rotor Angle with reference to the reference machine

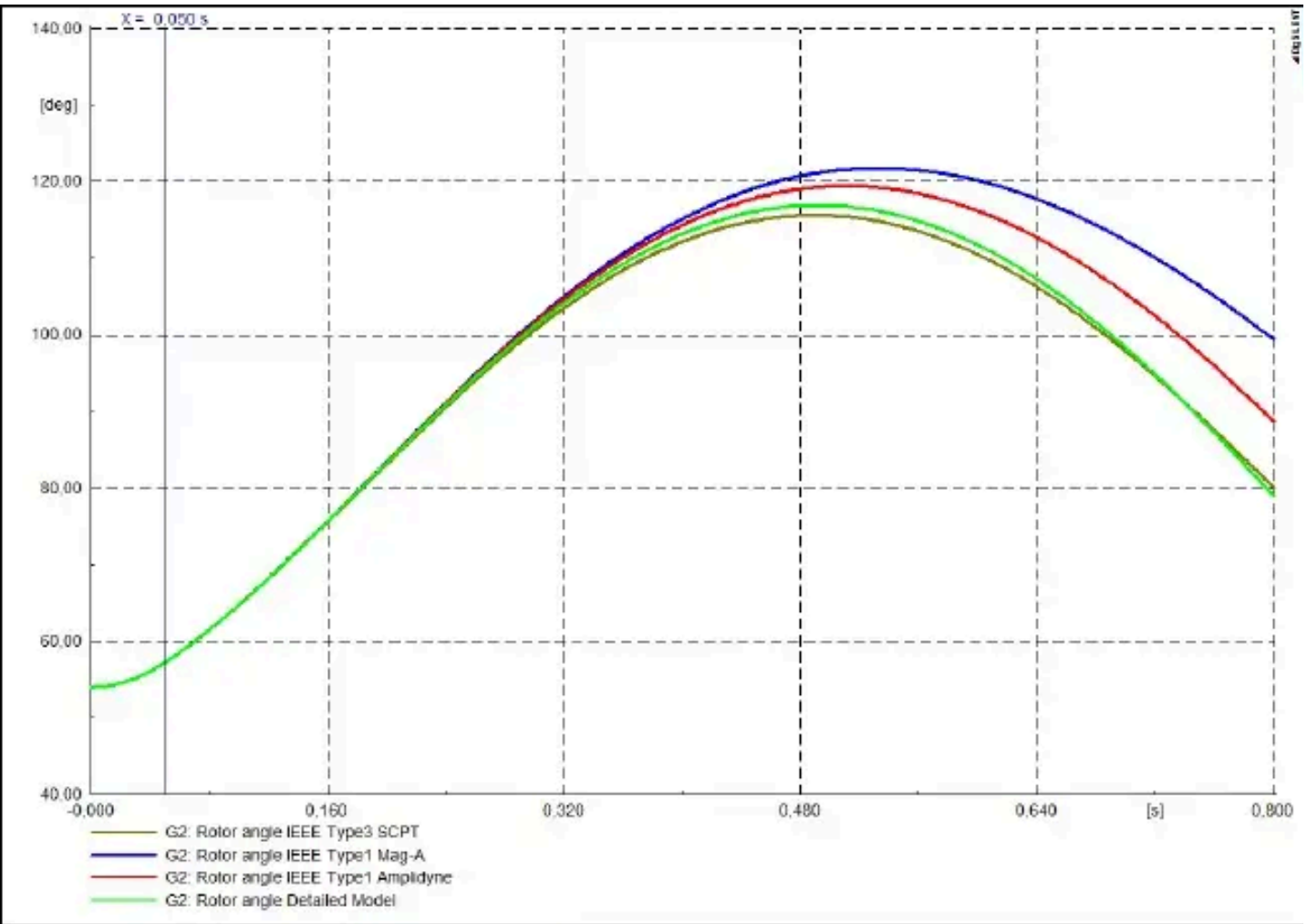


Figure 4: Comparison of the rotor angle for different excitation systems



Figure 5: Comparison of the rotor angle with and without PSS



Figure 6: Comparison of the excitation voltage with and without PSS

A Tables with Input Data

Table 1: Load demand [1]

Load	Bus	P [MW]	Q [Mvar]
Load A	Bus 5	125	50
Load B	Bus 6	90	30
Load C	Bus 8	100	35

Table 2: Generator Data (x based on 100 MVA) [1]

Quantity	G1	G2	G3
Nominal apparent power [MVA]	247.5	192.0	128.0
Nominal voltage [kV]	16.5	18.0	13.8
Nominal power factor	1.00	0.85	0.85
Type	hydro	steam	steam
Nominal speed [rpm]	180	3600	3600
x_d [p.u.]	0.1460	0.8958	1.3125
x'_d [p.u.]	0.0608	0.1198	0.1813
x_q [p.u.]	0.0969	0.8645	1.2578
x'_q [p.u.]	0.0969	0.1969	0.2500
x_l (leakage) [p.u.]	0.0336	0.0521	0.0742
τ'_{d0} [s]	8.960	6.000	5.890
τ'_{q0} [s]	0.000	0.535	0.600
Stored energy at nominal speed [MW · s]	2364	640	301

Table 3: Generator Data in the *PowerFactory* model (x based on rated power)

Quantity	G1	G2	G3
Nominal apparent power [MVA]	247.5	192.0	128.0
Nominal voltage [kV]	16.5	18.0	13.8
Nominal power factor	1.00	0.85	0.85
Plant Category	Hydro	Coil	Coil
Rotor Type	salient pole	round rotor	round rotor
x_d [p.u.]	0.3614	1.7199	1.6800
x'_d [p.u.]	0.1505	0.2300	0.2321
x_q [p.u.]	0.2328	1.6598	1.6100
x'_q [p.u.]	-	0.3780	0.3200
x_l (leakage) [p.u.]	0.0832	0.1000	0.0950
τ'_{d0} [s]	8.960	6.000	5.890
τ'_{q0} [s]	-	0.535	0.600
Inertia Constant H (Rated to Pgn) [s]	9.5515	3.9216	2.7665

Table 4: Generator dispatch and voltage setpoints [1]

Generator	Bus	P [MW]	u [p.u.]
G1	Bus 1	N/A	1.040
G2	Bus 2	163.0	1.025
G3	Bus 3	85	1.025

Table 5: Data of lines based on 100 MVA [1]

From	To	r [p.u.]	x [p.u.]	b/2 [p.u.]
Bus 4	Bus 5	0.0100	0.0850	0.0880
Bus 4	Bus 6	0.0170	0.0920	0.0790
Bus 5	Bus 7	0.0320	0.1610	0.1530
Bus 6	Bus 9	0.0390	0.1700	0.1790
Bus 7	Bus 8	0.0085	0.0720	0.0745
Bus 8	Bus 9	0.0119	0.1008	0.1045

Table 6: Data of lines in the *PowerFactory* model

Line	From	To	R [Ω]	X [Ω]	B [μS]
Line 4-5	Bus 4	Bus 5	5.2900	44.9650	332.70
Line 4-6	Bus 4	Bus 6	8.9930	48.6680	298.69
Line 5-7	Bus 5	Bus 7	16.928	85.1690	578.45
Line 6-9	Bus 6	Bus 9	20.631	89.9300	676.75
Line 7-8	Bus 7	Bus 8	4.4965	38.0880	281.66
Line 8-9	Bus 8	Bus 9	6.2951	53.3232	395.08

Table 7: Data of transformers based on 100 MVA [1]

Transformer	From	To	Ur HV [kV]	Ur LV [kV]	x1 [p.u.]
T1	Bus 1	Bus 4	230	16.5	0.0576
T2	Bus 2	Bus 7	230	18.0	0.0625
T3	Bus 3	Bus 9	230	13.8	0.0586

Table 8: Data of transformers in the *PowerFactory* model

Transformer	From	To	Rated Power [MVA]	Ur HV [kV]	Ur LV [kV]	x1 [p.u.]
T1	Bus 1	Bus 4	250	230	16.5	0.1440
T2	Bus 2	Bus 7	200	230	18.0	0.1250
T3	Bus 3	Bus 9	150	230	13.8	0.0879

B Tables with Results of the Balanced Load Flow Calculation

Table 9: Results of bus voltages provided in [1]

Name	u, Magnitude [p.u.]	u, Angle [deg]
Bus 1	1.040	0.0
Bus 2	1.025	9.3
Bus 3	1.025	4.7
Bus 4	1.026	-2.2
Bus 5	0.996	-4.0
Bus 6	1.013	-3.7
Bus 7	1.026	3.7
Bus 8	1.016	0.7
Bus 9	1.032	2.0

Table 10: Results of bus voltages obtained with *PowerFactory*

Name	U, Magnitude (line-line) [kV]	u, Magnitude [p.u.]	u, Angle (line-earth) [deg]
Bus 1	17.16	1.040	0.00
Bus 2	18.45	1.025	9.25
Bus 3	14.15	1.025	4.64
Bus 4	235.96	1.025	147.78 = 150.00 - 2.22
Bus 5	229.07	0.996	146.02 = 150.00 - 3.98
Bus 6	232.95	1.013	146.31 = 150.00 - 3.69
Bus 7	235.97	1.026	153.69 = 150.00 + 3.69
Bus 8	233.69	1.016	150.70 = 150.00 + 0.70
Bus 9	237.48	1.033	151.95 = 150.00 + 1.95

Table 11: Results of generators provided in [1]

Name	Active Power [MW]	Reactive Power [Mvar]
G1	71.6	27.0
G2	163.0	6.7
G3	85.0	-10.9

Table 12: Results of generators obtained with *PowerFactory*

Name	Active Power [MW]	Reactive Power [Mvar]
G1	71.60	26.78
G2	163.00	6.70
G3	85.00	-10.90

Table 13: Results of lines

Name	Losses [MW]	Reactive Losses [Mvar]	Capacitive Loading [Mvar]
Line 4-5	0.2551	-15.8229	17.9913
Line 4-6	0.1675	-15.5132	16.4196
Line 5-7	2.2969	-19.8453	31.4014
Line 6-9	1.3477	-31.5696	37.4443
Line 7-8	0.4735	-11.5217	15.5328
Line 8-9	0.0885	-21.1783	21.9284



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